Collectively Big, Locally Small: Online Supplement

Systematic search and coded corpus

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This supplement accompanies the survey *Collectively Big, Locally Small*. Section S1 documents the systematic database search summarized in Section 5.4 of the paper: sources, query strings, screening rules and their verification, the recall of the search against the coded corpus, the validation of the dictionary coder, and the field-level map. Section S2 lists the 290 coded works. The simulation study of the Graph Hierarchical Borrowing blueprint summarized in Section 8.7, its revision and the extensions tried are reported in full in the companion paper [Simões et al. 2026] and its supplement. Every number in this supplement is generated by the scripts released with the paper (`prisma/`) and read into the text as LaTeX macros.

S1 SYSTEMATIC SEARCH

S1.1 Sources and query strings

The search was run on 8 October 2026 for records published from 1 January 2010 to the search date. PubMed was queried through the NCBI E-utilities with every term restricted to title and abstract (`[tiab]`). medRxiv and the arXiv records indexed by Europe PMC were queried through the Europe PMC REST API (source `PPR`, field `TITLE_ABS`). Europe PMC indexes only the life-science categories of arXiv. A direct search of the arXiv API was attempted but abandoned because the API rejected the full query; computer-science preprints are therefore covered only when they were also published in an indexed venue. Every query string is stored verbatim in `prisma/data/search_log.json` together with the hit counts; the PubMed string was

```
("small data"[tiab] OR "small sample*"[tiab] OR "limited data"[tiab] OR "data scarcity"[tiab] OR
"few-shot"[tiab] OR "low-resource"[tiab] OR "rare disease*"[tiab] OR "label-efficient"[tiab] OR
"data-efficient"[tiab] OR federated[tiab] OR decentralized[tiab] OR decentralised[tiab] OR
"privacy-preserving"[tiab]) AND (health*[tiab] OR clinical[tiab] OR medic*[tiab] OR
hospital*[tiab] OR patient*[tiab]) AND ("machine learning"[tiab] OR "deep learning"[tiab] OR
"transfer learning"[tiab] OR "synthetic data"[tiab] OR "data augmentation"[tiab] OR "active
learning"[tiab] OR "semi-supervised"[tiab] OR "self-supervised"[tiab] OR "foundation
model*"[tiab] OR "hierarchical model*"[tiab]) AND 2010/01/01:2026/10/08[dp]
```

Scopus, Web of Science, IEEE Xplore and the ACM Digital Library were not searched, because they require a subscription or an API key. Exports from them (RIS, BibTeX or CSV) can be added with `prisma/import_exports.py`; they are then deduplicated and screened with the other records, and the flow diagram and every number are regenerated.

S1.2 Screening rules

Records were deduplicated by DOI and by normalized title, keeping the PubMed version when a preprint had also been published. Titles and abstracts were then screened by the rules of Table S1, applied in order. The rules operationalize criteria I1, I2 and X1 to X3 of Section 5: they keep every record in which a scarcity term of the query refers to scarce, fragmented or poorly labeled data and a learning method is present, and they remove the frequent uses of the query terms that do not refer to data scarcity.

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Table S1. Screening rules, applied in order; the first rule that applies decides.

Rule	Excludes records whose	*n*
A1	abstract is missing, or language is not English	46
A2	publication type is editorial, letter, comment, erratum, guideline, consensus statement, protocol, audio-visual material or historical article, or which are retracted (X1)	54
R1	only data-scarcity term refers to a resource-limited setting ("low-resource settings, countries, hospitals")	421
R2	only data-scarcity term is "small sample" in a sentence on limitations or future work	264
R3	only decentralization term refers to decentralized clinical trials or care delivery	14
R4	title is a reporting guideline, checklist, statement, position paper, policy or consensus (X1)	8
R5	title and abstract contain no learning method	72
	Eligible on title and abstract	5,204

S1.3 Verification by a single reviewer

A single reviewer judged two random samples of 100 records each against criteria I1, I2 and X1 to X3, reading title and abstract; each sample drew 60 or 40 records from the included side and as many from the excluded side of the rules then in use. The first sample led to the removal of one rule, which excluded records whose scarcity term appears only in the author keywords (4 of its 8 false exclusions came from it). A rule added after the first sample, which excluded "limited data" in the sense of a gap in clinical knowledge, removed small-training-set studies in 3 of the 7 sampled records it excluded in the second sample and was also removed. Against all 200 judgments, the final rules have a precision of 63% and a negative predictive value of 94%; the sensitivity within the samples is 94%, but the samples are balanced by design, so this is not the sensitivity in the full set. A stricter rule, which required the scarcity term in the title or in a sentence that also mentions data, labels, training or a model, raised precision to 73% but lowered sensitivity to 77% and was not adopted. The judgments are released in `prisma/data/verify_judgments.json`.

S1.4 Recall against the coded corpus

A search is usually validated by its relative recall against a reference set of known relevant studies. Our reference set is the coded corpus: 201 of its works were published since 2010 and evaluated on health data, 144 of these are indexed in the searched sources (found in Europe PMC by DOI or title), and the protocol query retrieved 31 of them (22%). To test whether a better query would have found them, we split the 140 PubMed-indexed works at random into two halves and evaluated broadened variants of the query on PubMed (Table S2). Adding 39 terms to the scarcity block (sample size, multi-site, external validation, generalizability, shift, privacy, annotation and related terms) and 16 terms to the method block (artificial intelligence, neural networks, prediction models and related terms) multiplied the number of records by thirteen and raised recall to about one half. The literature on learning from scarce health data is not well delimited by vocabulary. A large part of it reports a method or a failure without naming scarcity at all, which is why the paper's synthesis rests on the snowballed corpus.

S1.5 Dictionary coder and field-level map

Each eligible record was coded from its title and abstract by dictionaries for the six provenance channels, the ten scarcity classes and the data modality (`prisma/classify.py`). The primary channel is the channel whose terms match most often, with title matches counted twice. On the 147 corpus works whose abstracts are available in Europe PMC, the dictionary coder agreed with the manual primary channel in 72% of cases (Cohen's $\kappa = 0.65$) and recovered 63% of the manually

Table S2. Number of PubMed records and relative recall on two random halves of the PubMed-indexed corpus works, for the protocol query and for broadened variants.

Query	PubMed records	Recall, first half	Recall, second half
Protocol query (Section 5.4)	5,885	17%	27%
Scarcity block broadened	40,584	39%	44%
Method block broadened	11,178	21%	31%
Both broadened	75,983	51%	54%

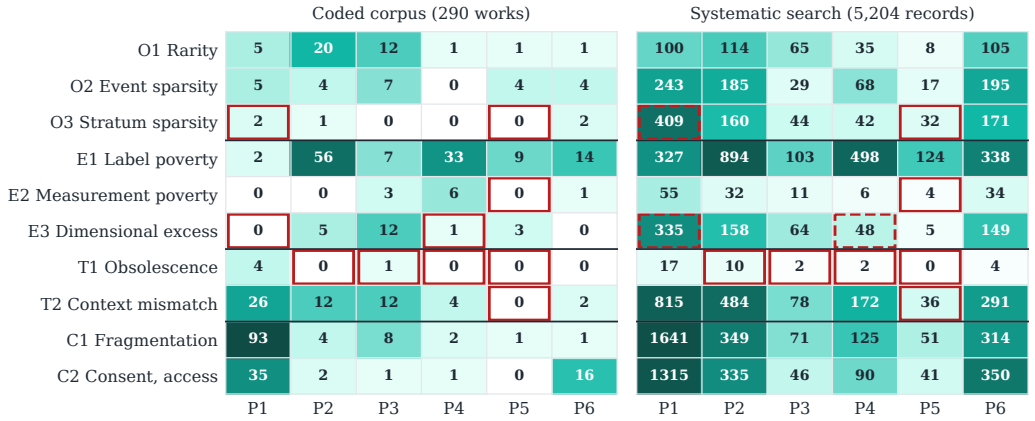


Fig. S1. Left: works of the coded corpus targeting each scarcity class through each channel, as in Figure 9 of the paper; red outlines mark the ten gap cells. Right: records of the systematic search coded by dictionary, shaded by percentile rank among the 60 cells. Solid outlines: gaps confirmed (bottom third of the field-level map); dashed outlines: gaps not confirmed.

assigned scarcity classes. Figure S1 sets the coverage of the coded corpus (Figure 9 of the paper) next to the field-level map. Because the dictionaries for stratum sparsity (which includes the term bias) and for dimensional excess (which includes omics terms) are broad, their rows are inflated in the field-level map; Section 5.4 of the paper discusses the three gap cells that the search does not confirm.

S2 THE CODED CORPUS

Table S3 lists all 290 works of the corpus with their coding under the scheme of Table 5 of the paper. The same data are released as a CSV file, together with the scripts that compute every count in Sections 5 and 7 and draw Figures 8 and 9.

Table S3. The coded corpus (290 works), grouped by primary provenance channel and sorted by year. Sec.: secondary channel. Scarcity: classes of Figure 3 the work targets or documents. Type: Method, Theory, Study (single-dataset empirical study), Multi-site (training or validation across institutions), Resource (dataset, benchmark or infrastructure), Review, Evidence (documented failure or risk). General: method not specific to health data.

Work	Modality	Sec.	Scarcity	Type
P1 Pooling (102)				
DerSimonian and Laird [1986]	Tabular	P3	C1, T2	Method

Work	Modality	Sec.	Scarcity	Type
Sweeney [2002]	Tabular		C2	Method
Xiao and Boyd [2004]	General		C1	Theory
Boyd et al. [2006]	General		C1	Theory
Dwork et al. [2006]	General		C2	Theory
Gentry [2009]	General		C2	Theory
Nedić and Ozdaglar [2009]	General		C1	Theory
Riley et al. [2010]	Tabular		C1, O2	Review
Overhage et al. [2012]	EHR		C2	Resource
Wu et al. [2012]	Tabular		C1	Method
Jiang et al. [2013]	Tabular		C1	Resource
Gaye et al. [2014]	Tabular		C1, C2	Resource
Hripcsak et al. [2015]	EHR		C1, C2	Resource
Lu et al. [2015]	Tabular		C1	Resource
Abadi et al. [2016]	General		C2	Method
Jochems et al. [2016]	Tabular		C1	Multi-site
Johnson et al. [2016]	EHR		C1, C2	Resource
Mandel et al. [2016]	EHR		C2	Resource
Blanchard et al. [2017]	General		C1	Method
Bonawitz et al. [2017]	General		C1, C2	Method
Jochems et al. [2017]	Tabular		C1	Multi-site
Kim et al. [2017]	EHR		C1	Method
Lian et al. [2017]	General		C1	Theory
McMahan et al. [2017]	General		C1	Method
Mironov [2017]	General		C2	Theory
Papernot et al. [2017]	General	P4	C2	Method
Smith et al. [2017]	General	P3	C1, T2	Method
Vanhaesebrouck et al. [2017]	General	P3	C1, T2	Method
Bellet et al. [2018]	General	P3	C1, T2	Method
Brisimi et al. [2018]	EHR		C1	Method
Chang et al. [2018]	Imaging		C1	Multi-site
Cho et al. [2018]	Omics		C1, C2	Method
Gupta and Raskar [2018]	General		C1	Method
Lee et al. [2018]	EHR		C1, C2	Method
McMahan et al. [2018]	General		C1, C2	Method
Reps et al. [2018]	EHR		C2	Resource
Vepakomma et al. [2018]	Multimodal		C1, C2	Method
Zhao et al. [2018]	General		C1, T2	Method
Choudhury et al. [2019]	EHR		C1, C2	Method
Huang et al. [2019]	EHR		C1, T2	Multi-site
Jordan et al. [2019]	General		C1	Theory
Li et al. [2019]	Imaging		C1, C2	Method
Mohri et al. [2019]	General		C1, O3	Method
Poirot et al. [2019]	Imaging		C1, C2	Method
Raisaro et al. [2019]	Multimodal		C1, C2	Method
Roy et al. [2019]	Imaging		C1	Method
Sheller et al. [2019]	Imaging		C1	Method
Suchard et al. [2019]	EHR		C1, C2, O2	Multi-site
Yang et al. [2019]	General		C1	Review
Beyan et al. [2020]	Multimodal		C1, C2	Resource
Chen et al. [2020b]	Signals	P2	C1, T2	Method
Deist et al. [2020]	Tabular		C1	Multi-site
T. Dinh et al. [2020]	General	P3	C1, T2	Method
Duan et al. [2020b]	EHR		C1, O2	Method
Duan et al. [2020a]	EHR		C1, O2	Method
Fallah et al. [2020]	General	P2	C1, T2	Method
Ghosh et al. [2020]	General		C1, T2	Method
Hsieh et al. [2020]	General		C1, T2	Method
Kaissis et al. [2020]	Imaging		C1, C2	Review
Karimireddy et al. [2020]	General		C1, T2	Method
Koloskova et al. [2020]	General		C1	Theory
Li et al. [2020c]	General		C1, T2	Method
Li et al. [2020a]	Imaging	P2	C1, T2	Multi-site
Lin et al. [2020]	General	P4	C1, T2	Method
Rieke et al. [2020]	Multimodal		C1, C2	Review
Roth et al. [2020]	Imaging		C1, T2	Multi-site
Sheller et al. [2020]	Imaging		C1, O1	Multi-site
Wei et al. [2020]	General		C1, C2	Method
Bai et al. [2021]	Imaging		C1, T1	Multi-site
Collins et al. [2021]	General	P2	C1, T2	Method
Dayan et al. [2021]	Multimodal		C1, T1, O2	Multi-site
Dou et al. [2021]	Imaging		C1, T1	Multi-site
Froelicher et al. [2021]	Omics		C1, C2	Method
Guo et al. [2021]	Imaging		C1	Method

Work	Modality	Sec.	Scarcity	Type
Huang et al. [2021a]	General		C1, T2	Method
Kairouz et al. [2021]	General		C1	Review
Kaissis et al. [2021]	Imaging		C1, C2	Method
Li et al. [2021a]	General	P3	C1, T2	Method
Li et al. [2021b]	General		C1, T2	Method
Liu et al. [2021]	Imaging	P6	C1, T2	Method
Marfoq et al. [2021]	General		C1, T2	Method
Sarma et al. [2021]	Imaging		C1	Multi-site
Tramèr and Boneh [2021]	General	P2	C2	Method
Vaid et al. [2021]	EHR		C1, T1	Multi-site
Warnat-Herresthal et al. [2021]	Multimodal		C1, O3	Multi-site
Xu et al. [2021]	Multimodal		C1	Review
Ziller et al. [2021]	Imaging		C2	Method
Adnan et al. [2022]	Imaging		C1, C2	Method
Cyffers and Bellet [2022]	General		C1, C2	Theory
Jiang et al. [2022]	Imaging		C1, T2	Method
Lu et al. [2022]	Imaging		C1, E1	Method
Nguyen et al. [2022]	Multimodal		C1	Review
Ogier du Terrail et al. [2022]	Multimodal		C1, T2	Resource
Pati et al. [2022]	Imaging		C1, O1	Multi-site
Saldanha et al. [2022]	Imaging		C1	Multi-site
Tedeschini et al. [2022]	Imaging		C1	Method
Xu et al. [2022]	Imaging		C1, T2	Method
Kalra et al. [2023]	Imaging		C1, C2	Method
Lu et al. [2023]	General		C1	Method
Ogier du Terrail et al. [2023]	Imaging		C1, O1	Multi-site
Tan et al. [2023]	General		C1, T2	Review
Yan et al. [2023]	Imaging	P4	C1, E1	Method
P2 Proxy (60)				
Baxter [2000]	General		O1, E1	Theory
Ben-David et al. [2010]	General		T2	Theory
Pan and Yang [2010]	General		E1, T2	Review
Choi et al. [2016]	EHR		E1, T2	Method
Gulshan et al. [2016]	Imaging		E1	Study
Maurer et al. [2016]	General		O1	Theory
Shin et al. [2016]	Imaging		E1	Study
Tajbakhsh et al. [2016]	Imaging		E1	Study
Esteva et al. [2017]	Imaging		E1	Study
Finn et al. [2017]	General		O1, E1	Method
Kamnitsas et al. [2017]	Imaging	P4	T2	Method
Snell et al. [2017]	General		O1, E1	Method
Alsentzer et al. [2019]	Text		E1	Method
Chen et al. [2019]	Imaging		E1	Method
Gurovich et al. [2019]	Imaging		O1	Method
Prabhu et al. [2019]	Imaging		O1, O3	Method
Raghu et al. [2019]	Imaging		E1	Study
Zhang et al. [2019]	EHR		O1, E1	Method
Brown et al. [2020]	General		E1	Method
Lee et al. [2020]	Text		E1	Method
Li et al. [2020b]	EHR		E1, E3	Method
Mahajan et al. [2020]	Imaging		O1	Method
Roy et al. [2020]	Imaging		E1, O1	Method
Tripuraneni et al. [2020]	General		E1	Theory
Wang et al. [2020]	General		O1, E1	Review
Azizi et al. [2021]	Imaging	P4	E1, T2	Method
Du et al. [2021]	General		E1	Theory
Huang et al. [2021b]	Imaging	P4	E1	Method
Radford et al. [2021]	General		E1	Method
Rasmy et al. [2021]	EHR		E1, O2	Method
Sowrirajan et al. [2021]	Imaging	P4	E1	Method
Steinberg et al. [2021]	EHR		E1	Method
Zhou et al. [2021]	Imaging	P4	E1	Method
Agrawal et al. [2022]	Text		E1	Method
Guan and Liu [2022]	Imaging		T2	Review
Hospedales et al. [2022]	General		O1, E1	Review
Hsieh et al. [2022]	Imaging		O1	Method
Hu et al. [2022]	General		E1	Method
Mei et al. [2022]	Imaging		E1	Resource
Tiu et al. [2022]	Imaging	P4	E1	Method
Wang et al. [2022]	Imaging	P4	E1	Method
Zhang et al. [2022]	Imaging	P4	E1	Method
Banerjee et al. [2023]	Multimodal	P3	O1	Review

Work	Modality	Sec.	Scarcity	Type
Hegselmann et al. [2023]	Tabular	P3	E1, O2	Method
Hollmann et al. [2023]	Tabular	P3	E3, O2	Method
Jiang et al. [2023]	Text		E1	Method
Moor et al. [2023]	Multimodal		E1, O1	Review
Nori et al. [2023]	Text		E1	Study
Singhal et al. [2023]	Text		E1	Method
Theodoris et al. [2023]	Omics		E1, E3, O1	Method
Wornow et al. [2023]	EHR		E1, T2	Review
Zhou et al. [2023]	Imaging	P4	E1, T2	Method
Chen et al. [2024]	Imaging		E1, O1	Method
Cui et al. [2024]	Omics		E1, E3	Method
Lu et al. [2024]	Imaging		E1, O1	Method
Ma et al. [2024]	Imaging		E1	Method
Pai et al. [2024]	Imaging		E1, O1	Method
Vorontsov et al. [2024]	Imaging		E1, O1	Method
Xu et al. [2024]	Imaging		E1	Method
Hollmann et al. [2025]	Tabular	P3	E3, O2	Method
P3 Prior (25)				
Stein [1956]	General	P1	O1, C1	Theory
Efron and Morris [1973]	General	P1	O1, C1	Theory
Tibshirani [1996]	General		E3	Method
Steyerberg et al. [2000]	Tabular		O2, E3	Study
Lake et al. [2015]	General		O1	Method
Choi et al. [2017a]	EHR		E3, O1	Method
Rotmensch et al. [2017]	EHR	P4	E3	Method
Ma et al. [2018]	EHR		E3, O1	Method
Arjovsky et al. [2019]	General		T1, T2	Method
Raissi et al. [2019]	General		O1, E3	Method
Shang et al. [2019]	EHR		E3	Method
Subbaswamy et al. [2019]	EHR		T2	Method
Castro et al. [2020]	Imaging		T2, E2	Review
Kissas et al. [2020]	Imaging		E1	Method
Lagergren et al. [2020]	Tabular		E1, E3	Method
Sahli Costabal et al. [2020]	Signals		O1, E1	Method
Van Calster et al. [2020]	Tabular		O2	Study
Gulrajani and Lopez-Paz [2021]	General		T2	Study
Karniadakis et al. [2021]	General		O1, E3	Review
Köhler et al. [2021]	Text		O1	Resource
Lu et al. [2021a]	Tabular		O1	Method
Riley et al. [2021]	Tabular		O2	Study
Schölkopf et al. [2021]	General		T2	Review
Zhang et al. [2021]	Multimodal		T2	Study
von Rueden et al. [2023]	General		O1, E3	Review
P4 Unlabeled (23)				
Schölkopf et al. [2012]	General		E1	Theory
Agarwal et al. [2016]	EHR		E1, E2	Method
Halpern et al. [2016]	EHR	P3	E1, E2	Method
Bai et al. [2017]	Imaging		E1	Method
Ratner et al. [2017]	General	P3	E1, E2	Method
Wang et al. [2017]	Imaging		E1, E2	Resource
Ilse et al. [2018]	Imaging		E1	Method
Campanella et al. [2019]	Imaging		E1	Method
Cheplygina et al. [2019]	Imaging	P2	E1	Review
Fries et al. [2019]	Imaging		E1	Method
Irvin et al. [2019]	Imaging		E1, E2	Resource
Wang et al. [2019]	Text	P3	E1	Method
Yu et al. [2019]	Imaging		E1	Method
Chaitanya et al. [2020]	Imaging	P2	E1	Method
Chen et al. [2020a]	General	P2	E1	Method
Ouyang et al. [2020]	Imaging	P2	E1, O1	Method
Sohn et al. [2020]	General		E1	Method
Tajbakhsh et al. [2020]	Imaging	P6	E1, E2	Review
Taleb et al. [2020]	Imaging	P2	E1	Method
Li et al. [2021c]	Imaging		E1	Method
Lu et al. [2021b]	Imaging		E1	Method
He et al. [2022]	General	P2	E1	Method
Krishnan et al. [2022]	Multimodal	P2	E1	Review
P5 Procurement (15)				
Peduzzi et al. [1996]	Tabular		O2	Theory
Settles [2009]	General		E1	Review
Figueroa et al. [2012]	Text		E1	Method

Work	Modality	Sec.	Scarcity	Type
Cho et al. [2015]	Imaging		E1	Study
Gal et al. [2017]	General		E1	Method
Yang et al. [2017]	Imaging		E1	Method
Smailagic et al. [2018]	Imaging		E1	Method
Balki et al. [2019]	Imaging		E1, O1	Review
Ghorbani and Zou [2019]	General		C1	Method
Riley et al. [2019b]	Tabular		O2, E3	Theory
Riley et al. [2019a]	Tabular		E3	Theory
van Smeden et al. [2019]	Tabular	P3	O2	Theory
Riley et al. [2020]	Tabular		O2, E3	Theory
Budd et al. [2021]	Imaging		E1	Review
Nath et al. [2021]	Imaging		E1	Method
P6 Re-expression (32)				
Chawla et al. [2002]	Tabular		O2	Method
Goodfellow et al. [2014]	General		E1	Method
Choi et al. [2017b]	EHR		C2	Method
Bowles et al. [2018]	Imaging		E1	Method
Fernández et al. [2018]	Tabular		O2	Review
Frid-Adar et al. [2018]	Imaging		E1, O2	Study
Shin et al. [2018]	Imaging		E1, C2	Method
Walonoski et al. [2018]	EHR	P3	C2	Resource
Zhang et al. [2018]	General		E1	Method
Baowaly et al. [2019]	EHR		C2	Method
Beaulieu-Jones et al. [2019]	Tabular	P1	C2	Method
Jordon et al. [2019]	Tabular	P1	C2	Method
Shorten and Khoshgoftaar [2019]	General		E1	Review
Xu et al. [2019]	Tabular		C2	Method
Yi et al. [2019]	Imaging		E1	Review
Zhao et al. [2019]	Imaging		E1	Method
Goncalves et al. [2020]	EHR		C2	Study
Ho et al. [2020]	General		E1	Method
Yoon et al. [2020]	EHR		C2	Method
Zhang et al. [2020]	EHR		C2	Method
Chaitanya et al. [2021]	Imaging	P4	E1	Method
Chen et al. [2021a]	Multimodal		C2, O1	Review
Chlap et al. [2021]	Imaging		E1	Review
Jordon et al. [2022]	General		C2	Review
Stadler et al. [2022]	Tabular		C2	Study
van den Goorbergh et al. [2022]	Tabular		O2	Study
Borisov et al. [2023]	Tabular	P2	C2	Method
Kazerouni et al. [2023]	Imaging		E1	Review
Khader et al. [2023]	Imaging		E1, C2	Method
Kotelnikov et al. [2023]	Tabular		C2	Method
Ktena et al. [2024]	Imaging		O3, T2	Method
Shumailov et al. [2024]	General		O3	Evidence
No remedy: documented scarcity, failure or risk (33)				
El Emam et al. [2011]	Tabular		C2	Review
Button et al. [2013]	General		O2	Evidence
Shokri et al. [2017]	General		C2	Evidence
Varoquaux [2018]	Imaging		O1, O2	Evidence
Zech et al. [2018]	Imaging		T2	Evidence
Bagdasaryan et al. [2019]	General		O3, C2	Evidence
Brasil et al. [2019]	Multimodal		O1	Review
Christodoulou et al. [2019]	Tabular		O2	Review
Jayaraman and Evans [2019]	General		C2	Evidence
Kim et al. [2019]	Imaging		T2	Review
Liu et al. [2019]	Imaging		T2	Review
Obermeyer et al. [2019]	EHR		O3	Evidence
Rocher et al. [2019]	Tabular		C2	Evidence
Schwarz et al. [2019]	Imaging		C2	Evidence
Zhu et al. [2019]	General		C1, C2	Evidence
Geiping et al. [2020]	General		C1, C2	Evidence
Nagendran et al. [2020]	Imaging		T2	Review
Nguengang Wakap et al. [2020]	Tabular		O1	Evidence
Oakden-Rayner et al. [2020]	Imaging		O3	Evidence
Schaefer et al. [2020]	Multimodal		O1	Review
Willemink et al. [2020]	Imaging		E1, E2, C2	Review
Wynants et al. [2020]	Multimodal		O2, T1	Evidence
Chen et al. [2021b]	Multimodal		O3	Review
Finlayson et al. [2021]	Multimodal		T1, T2	Evidence
Koh et al. [2021]	Multimodal		T2	Resource
Roberts et al. [2021]	Imaging		O2, T1	Evidence

Work	Modality	Sec.	Scarcity	Type
Wong et al. [2021]	EHR		T2	Evidence
Carlini et al. [2022]	General		C2	Evidence
Gichoya et al. [2022]	Imaging		O3, T2	Evidence
Packhäuser et al. [2022]	Imaging		C2	Evidence
Varoquaux and Cheplygina [2022]	Imaging		O1, T2	Evidence
Yu et al. [2022]	Imaging		T2	Review
Hatamizadeh et al. [2023]	Imaging		C1, C2	Evidence

S3 DATA AND REPORTED RESULTS OF EVERY CODED WORK

For each of the 206 corpus works evaluated on health data, this section gives the data it uses and its headline result (Section 7.2 of the paper). They were extracted from the work's Europe PMC abstract when available (`prisma/data/corpus_lookup.jsonl`) and otherwise from its arXiv abstract, its PubMed record or its own page. Every non-empty field is supported by a verbatim quote, stored with its source in `data/corpus_extraction.json`; for works taken from their Europe PMC abstract, `scripts/corpus_extract.py` checks that each quote occurs in the abstract or title. Where an abstract does not name its data, the entry describes the data in the abstract's words or reads "not stated"; "no number in abstract" means the abstract gives no headline number, not that the paper has none. Sizes and results are quoted as the authors state them. Works are grouped by primary channel and ordered by year.

P1 Pooling (62 works)

Work	Year	Modality	Data (size)	Reported result
[DerSimonian and Laird 1986]	1986	Tabular	no dataset (review, theory, guidance or simulation)	-
[Sweeney 2002]	2002	Tabular	no dataset (review, theory, guidance or simulation)	-
[Riley et al. 2010]	2010	Tabular	no dataset (review, theory, guidance or simulation)	-
[Overhage et al. 2012]	2012	EHR	10 observational healthcare databases	no number in abstract
[Wu et al. 2012]	2012	Tabular	simulated data; clinically relevant data (unnamed)	GLORE results identical to centrally estimated logistic regression to 10^-15 precision
[Jiang et al. 2013]	2013	Tabular	no dataset (review, theory, guidance or simulation)	-
[Gaye et al. 2014]	2014	Tabular	Healthy Obese Project and Environmental Core Project (BioSHaRE-EU) data sets (10 data sets)	no number in abstract
[Hripcsak et al. 2015]	2015	EHR	no dataset (review, theory, guidance or simulation)	-
[Lu et al. 2015]	2015	Tabular	publicly available dataset (unnamed), 3 sites	no number in abstract
[Jochems et al. 2016]	2016	Tabular	Lung cancer patients from MAASTRO, Jessa, Liege, Aachen and Eindhoven (287 patients), 5 sites	AUC 0.61 (95%CI 0.51-0.70) in 5-fold cross-validation; 0.59 to 0.71 on external validation sets
[Johnson et al. 2016]	2016	EHR	MIMIC-III, 1 site	no number in abstract
[Mandel et al. 2016]	2016	EHR	no dataset (review, theory, guidance or simulation)	-
[Jochems et al. 2017]	2017	Tabular	Maastro clinic (Netherlands) cohort (559 patients); Michigan university (United States) cohort (139 patients); The Christie (United Kingdom) external validation cohort (196 patients)	AUC 0.66 on external validation set and 0.62 in 5-fold cross-validation; T and N category model AUC 0.47 on validation set.
[Kim et al. 2017]	2017	EHR	real medical datasets (multiple hospitals; not named)	no number in abstract
[Brisimi et al. 2018]	2018	EHR	patients' Electronic Health Records (unnamed)	no number in abstract
[Chang et al. 2018]	2018	Imaging	Kaggle Diabetic Retinopathy (retinal fundus photos) (35 126 color digital retinal fundus photos); Curated Breast Imaging Subset of the Digital Database for Screening Mammography (DDSM); ImageNet	no number in abstract
[Cho et al. 2018]	2018	Omics	genotype/phenotype cohorts (unnamed) (9K, 13K, and 23K individuals)	no number in abstract

Work	Year	Modality	Data (size)	Reported result
[Lee et al. 2018]	2018	EHR	MIMIC-III	Averaged AUC 0.9154 (balanced) and 0.8012 (imbalanced data) with k-nearest neighbor, k=3
[Reps et al. 2018]	2018	EHR	4 observational databases (OMOP Common Data Model)	no number in abstract
[Vepakomma et al. 2018]	2018	Multimodal	not stated	no number in abstract
[Choudhury et al. 2019]	2019	EHR	Limited MarketScan Explorys Claims-EMR Data (LCED) (over 1 million commercially insured patients); MIMIC III (over 40,000 ICU stays)	no number in abstract
[Huang et al. 2019]	2019	EHR	non-IID ICU EMRs	no number in abstract
[Li et al. 2019]	2019	Imaging	BraTS	no number in abstract
[Poirot et al. 2019]	2019	Imaging	fundus photo data set (9000 fundus photos); chest X-ray data set (156,535 chest X-rays)	Split learning performance constant for 1-50 clients vs single-center; marked difference vs non-collaborative after 2 clients ($p < 0.001$)
[Raisaro et al. 2019]	2019	Multimodal	The Cancer Genome Atlas	no number in abstract
[Roy et al. 2019]	2019	Imaging	not stated	no number in abstract
[Sheller et al. 2019]	2019	Imaging	multimodal brain scans (multi-institutional, unnamed)	Federated segmentation Dice 0.852 vs 0.862 for models trained by sharing data
[Suchard et al. 2019]	2019	EHR	global network of six administrative claims and three electronic health record databases (4·9 million patients)	Thiazide diuretics vs ACE inhibitors: acute MI HR 0·84 (0·75-0·95), heart failure hospitalisation 0·83, stroke 0·83
[Beyan et al. 2020]	2020	Multimodal	abstract not accessible	-
[Chen et al. 2020b]	2020	Signals	not stated	5.3% accuracy improvement for wearable activity recognition compared to traditional methods
[Deist et al. 2020]	2020	Tabular	Lung cancer patient databases of oncology departments (Personal Health Train) (23 203 patient cases), 8 sites	no number in abstract
[Duan et al. 2020a]	2020	EHR	dataset from the University of Pennsylvania Health System, 1 site	no number in abstract
[Duan et al. 2020b]	2020	EHR	4 datasets from the Observational Health Data Sciences and Informatics (OHDSI) network; simulation study	OHDSI application: ODAC relative bias <5% for 15 of 16 log hazard ratios; meta-analysis estimates had substantially higher bias. Simulation: relative bias <0.1% vs pooled estimator.
[Kaissis et al. 2020]	2020	Imaging	no dataset (review, theory, guidance or simulation)	-
[Li et al. 2020a]	2020	Imaging	ABIDE	no number in abstract
[Rieke et al. 2020]	2020	Multimodal	no dataset (review, theory, guidance or simulation)	-
[Roth et al. 2020]	2020	Imaging	Mammography datasets from seven clinical institutions, 7 sites	FL models 6.3% better on average than models trained on local data alone; 45.8% relative improvement in generalizability to other sites.
[Sheller et al. 2020]	2020	Imaging	multi-institutional data (10 federated institutions; not named in abstract), 10 sites	Federated learning among 10 institutions reached 99% of the model quality achieved with centralized data.
[Bai et al. 2021]	2021	Imaging	UCADI chest CT scans from hospitals in China and the UK (9,573 chest CT scans from 3,336 patients), 23 sites	FL model test sensitivity/specificity China 0.973/0.951, UK 0.730/0.942, outperforming all local models
[Dayan et al. 2021]	2021	Multimodal	data from 20 institutes across the globe (vital signs, laboratory data, chest X-rays), 20 sites	Average AUC >0.92 at 24 and 72 h; 16% average AUC improvement and 38% generalizability increase vs single-site models
[Dou et al. 2021]	2021	Imaging	multinational COVID-19 chest CT cohort (Hong Kong, Mainland China, Germany) (132 patients), 7 sites	no number in abstract
[Froelicher et al. 2021]	2021	Omics	not stated	no number in abstract
[Guo et al. 2021]	2021	Imaging	not stated	no number in abstract
[Kaissis et al. 2021]	2021	Imaging	abstract not accessible	-
[Liu et al. 2021]	2021	Imaging	retinal fundus images from public datasets, 4 sites; prostate T2-weighted MRI images from public datasets, 6 sites	Fundus segmentation: ELCFS improves over FedAvg by 2.02% Dice and 2.86 HD overall
[Sarma et al. 2021]	2021	Imaging	Private prostate MRI datasets from UCLA, SUNY Upstate Medical University and NCI (1 prostate MRI from each of 100 patients per institution), 3 sites; SPIE-AAPM-NCI PROSTATEx (challenge dataset) (343 axial T2W images)	FL model overall mean Dice 0.889, significantly superior to every single-institution model ($P < .001$)
[Vaid et al. 2021]	2021	EHR	Mount Sinai Health System EHR data, 5 sites	no number in abstract

Work	Year	Modality	Data (size)	Reported result
[Warnat-Herresthal et al. 2021]	2021	Multimodal	blood transcriptomes from 127 clinical studies (more than 16,400 blood transcriptomes); chest X-ray images (more than 95,000 chest X-ray images)	no number in abstract
[Xu et al. 2021]	2021	Multimodal	no dataset (review, theory, guidance or simulation)	-
[Ziller et al. 2021]	2021	Imaging	paediatric pneumonia dataset; Medical Segmentation Decathlon Liver dataset	no number in abstract
[Adnan et al. 2022]	2022	Imaging	The Cancer Genome Atlas (TCGA)	no number in abstract
[Jiang et al. 2022]	2022	Imaging	Camelyon17 (450,000 histology images), 5 sites; MoNuSAC2020; MoNuSAC2018; multi-site prostate segmentation dataset (6 data sources from 3 public datasets)	no number in abstract
[Lu et al. 2022]	2022	Imaging	histology whole slide images (unnamed) (thousands of histology whole slide images)	no number in abstract
[Nguyen et al. 2022]	2022	Multimodal	no dataset (review, theory, guidance or simulation)	-
[Ogier du Terrail et al. 2022]	2022	Multimodal	FLamby (7 healthcare datasets with natural splits) (7 datasets)	no number in abstract
[Pati et al. 2022]	2022	Imaging	glioblastoma data from 71 sites across 6 continents (n = 6,314), 71 sites	33% delineation improvement for surgically targetable tumor and 23% for complete tumor extent over a publicly trained model.
[Saldanha et al. 2022]	2022	Imaging	Epi700 (661 patients); DACHS (2,448 patients); TCGA (632 patients); QUASAR; YCR BCIP	MSI in QUASAR: swarm AUROCs 0.8001 and 0.8151 vs 0.7884 for Epi700 single-cohort model
[Tedeschini et al. 2022]	2022	Imaging	representative clinical datasets obtained from the daily clinical workflow	no number in abstract
[Xu et al. 2022]	2022	Imaging	not stated	no number in abstract
[Kalra et al. 2023]	2023	Imaging	popular image datasets; gigapixel histology whole slide images (cancer diagnostic problem)	no number in abstract
[Ogier du Terrail et al. 2023]	2023	Imaging	multicentric TNBC whole-slide image and clinical data	no number in abstract
[Yan et al. 2023]	2023	Imaging	simulated and real-world medical imaging non-IID federated datasets (retinal, dermatology, chest X-ray)	Under severe heterogeneity, test accuracy improved 5.06%, 1.53%, 4.58% (retinal, dermatology, chest X-ray) vs supervised baseline with ImageNet pre-training.

P2 Proxy (47 works)

Work	Year	Modality	Data (size)	Reported result
[Choi et al. 2016]	2016	EHR	longitudinal time stamped EHR data (unnamed) (260K patients over 8 years)	Differential diagnosis up to 79% recall@30, significantly higher than several baselines
[Gulshan et al. 2016]	2016	Imaging	retrospective development data set (128 175 retinal images); EyePACS-1 (9963 images from 4997 patients); Messidor-2 (1748 images from 874 patients)	AUC 0.991 (EyePACS-1) and 0.990 (Messidor-2) for referable diabetic retinopathy
[Shin et al. 2016]	2016	Imaging	not stated	no number in abstract
[Tajbakhsh et al. 2016]	2016	Imaging	not stated	no number in abstract
[Esteva et al. 2017]	2017	Imaging	clinical skin lesion images (129,450 clinical images)	no number in abstract
[Kamnitsas et al. 2017]	2017	Imaging	two MR databases of subjects with traumatic brain injuries	no number in abstract
[Alsentzer et al. 2019]	2019	Text	MIMIC-III v1.4 (approximately 2 million notes); MedNLI; i2b2 2006; i2b2 2010; i2b2 2012; i2b2 2014	MedNLI accuracy 82.7% vs prior state of the art 73.5%
[Chen et al. 2019]	2019	Imaging	3DSeg-8 (aggregated from several medical challenges); LIDC; LiTS challenge	2x faster convergence vs Kinetics pretraining, 10x vs scratch; accuracy gains 3% to 20%; 94.6% Dice on LiTS
[Gurovich et al. 2019]	2019	Imaging	community-driven phenotyping platform facial images (over 17,000 images)	DeepGestalt 91% top-10 accuracy identifying the correct syndrome on 502 images
[Prabhu et al. 2019]	2019	Imaging	realistic dataset of dermatological conditions	no number in abstract
[Raghu et al. 2019]	2019	Imaging	not stated	no number in abstract
[Zhang et al. 2019]	2019	EHR	real patient EHR repository from Oregon Health & Science University, 1 site	no number in abstract
[Lee et al. 2020]	2020	Text	PubMed abstracts (4.5B words); PubMed Central (PMC) full-text articles (13.5B words)	Vs previous SOTA: NER +0.62% F1, relation extraction +2.80% F1, QA +12.24% MRR

Work	Year	Modality	Data (size)	Reported result
[Li et al. 2020b]	2020	EHR	Clinical Practice Research Datalink (CPRD) (nearly 1.6 million individuals), 674 sites	8.0-13.2% improvement in average precision over state-of-the-art deep EHR models
[Mahajan et al. 2020]	2020	Imaging	abstract not accessible	-
[Roy et al. 2020]	2020	Imaging	Visceral Dataset	no number in abstract
[Azizi et al. 2021]	2021	Imaging	dermatology digital camera images; chest X-ray images (multi-label)	+6.7% top-1 accuracy (dermatology) and +1.1% mean AUC (chest X-ray) over supervised ImageNet-pretrained baselines.
[Huang et al. 2021b]	2021	Imaging	not stated	no number in abstract
[Rasmy et al. 2021]	2021	EHR	Cerner Health Facts (28,490,650 patients); Truven Health MarketScan	AUC improved by 1.21-6.14% in two disease prediction tasks; >20% with small fine-tuning sets
[Sowrirajan et al. 2021]	2021	Imaging	CheXpert (224k chest X-rays collected from 65k patients); Shenzhen Hospital X-ray set (662 X-ray images)	Pleural effusion, 0.1% labels: linear model AUC 0.776 with MoCo-CXR vs 0.683 ImageNet-pretrained (+0.096)
[Steinberg et al. 2021]	2021	EHR	not stated	3.5% mean AUROC improvement over standard baselines on five tasks; 19% average improvement with few training records
[Zhou et al. 2021]	2021	Imaging	not stated	no number in abstract
[Agrawal et al. 2022]	2022	Text	CASI dataset (re-annotated)	no number in abstract
[Guan and Liu 2022]	2022	Imaging	no dataset (review, theory, guidance or simulation)	-
[Hsieh et al. 2022]	2022	Imaging	patient photographs (Clinical Face Phenotype Space) (17,560 patients with 1,115 rare disorders)	no number in abstract
[Mei et al. 2022]	2022	Imaging	RadImageNet (1.35 million annotated medical images in 131 872 patients)	no number in abstract
[Tiu et al. 2022]	2022	Imaging	not stated	no number in abstract
[Wang et al. 2022]	2022	Imaging	not stated	With only 20K pre-training data, MedCLIP outperforms the state-of-the-art method using ≈200K data
[Zhang et al. 2022]	2022	Imaging	not stated	Needs only 10% as much labeled data as ImageNet-initialized counterpart for better or comparable performance on 4 classification tasks.
[Banerjee et al. 2023]	2023	Multimodal	no dataset (review, theory, guidance or simulation)	-
[Hegselmann et al. 2023]	2023	Tabular	Bank (45,211 rows); Blood (748 rows); California (20,640 rows); Car (1,728 rows); Credit-g (1,000 rows); Income (48,842 rows); Jungle (44,819 rows); Diabetes (Kaggle) (768 rows); Heart (Kaggle) (918 rows); de-identified health claims dataset from a U.S. health insurer	Zero-shot AUC on health claims tasks from 0.67 (Surgery) to 0.71 (LoH)
[Hollmann et al. 2023]	2023	Tabular	OpenML-CC18 suite (18 datasets) (up to 1 000 training data points each); additional small numerical datasets from OpenML (67 datasets)	Outperforms boosted trees, on par with state-of-the-art AutoML with up to 230x speedup (5 700x on GPU)
[Jiang et al. 2023]	2023	Text	clinical notes from the authors' health system	NYUTron AUC 78.7-94.9%, improving AUC by 5.36-14.7% over traditional models
[Moor et al. 2023]	2023	Multimodal	no dataset (review, theory, guidance or simulation)	-
[Nori et al. 2023]	2023	Text	USMLE official practice materials; MultiMedQA	GPT-4 exceeds USMLE passing score by over 20 points; outperforms GPT-3.5 and Med-PaLM.
[Singhal et al. 2023]	2023	Text	MultiMedQA; HealthSearchQA; MedQA; MedMCQA; PubMedQA; MMLU clinical topics	Flan-PaLM 67.6% accuracy on MedQA, surpassing prior state of the art by more than 17%.
[Theodoris et al. 2023]	2023	Omics	large-scale corpus of single-cell transcriptomes (unnamed) (about 30 million single-cell transcriptomes)	no number in abstract
[Wornow et al. 2023]	2023	EHR	no dataset (review, theory, guidance or simulation)	-
[Zhou et al. 2023]	2023	Imaging	unlabelled retinal images (RETFound pretraining) (1.6 million unlabelled retinal images)	no number in abstract
[Chen et al. 2024]	2024	Imaging	diagnostic H&E-stained WSIs across 20 major tissue types (pretraining) (more than 100 million images from over 100,000 WSIs)	no number in abstract
[Cui et al. 2024]	2024	Omics	single-cell sequencing data repository (over 33 million cells)	no number in abstract

Work	Year	Modality	Data (size)	Reported result
[Lu et al. 2024]	2024	Imaging	histopathology image-caption pairs from diverse sources (CONCH pretraining) (over 1.17 million image-caption pairs)	no number in abstract
[Ma et al. 2024]	2024	Imaging	curated dataset from publicly available medical image segmentation datasets (1,570,263 image-mask pairs)	no number in abstract
[Pai et al. 2024]	2024	Imaging	DeepLesion (11,467 radiographic lesions used for pretraining; dataset comprises 32,735 lesions from 4,427 unique patients); LUNA16 (507 lung nodules); HarvardRT (n = 291); LUNG1 (NSCLC-Radiomics) (n = 420); RADIO (NSCLC-Radiogenomics) (n = 133)	LUNG1 2-year survival: Foundation (features) AUC 0.638 (95% CI 0.584-0.692), exceeding all baselines
[Vorontsov et al. 2024]	2024	Imaging	not stated	Pan-cancer detection specimen-level AUC 0.95 across nine common and seven rare cancers
[Xu et al. 2024]	2024	Imaging	Providence health network whole slides (171,189 whole slides (1.3 billion tiles; more than 30,000 patients)), 28 sites; TCGA	State-of-the-art on 25 of 26 tasks; significant improvement over second-best method on 18 tasks
[Hollmann et al. 2025]	2025	Tabular	synthetic datasets (pretraining) (millions of synthetic datasets)	In 2.8 s, TabPFN outperforms an ensemble of strongest baselines tuned for 4 h (classification, datasets up to 10,000 samples).

P3 Prior (15 works)

Work	Year	Modality	Data (size)	Reported result
[Steyerberg et al. 2000]	2000	Tabular	small data sets of patients with acute myocardial infarction	no number in abstract
[Choi et al. 2017a]	2017	EHR	electronic health records (unnamed)	Vs basic RNN: 10% higher accuracy for rare diseases; 3% higher heart-failure AUC
[Rotmensch et al. 2017]	2017	EHR	de-identified patient records (273,174 de-identified patient records)	Noisy OR model precision 0.85 at recall 0.6 in clinical evaluation; outperforms logistic regression and naive Bayes ($p < 0.01$)
[Ma et al. 2018]	2018	EHR	abstract not accessible	-
[Shang et al. 2019]	2019	EHR	real EHR data	Outperformed all baselines in all effectiveness measures; 3.60% DDI rate reduction from existing EHR data
[Subbaswamy et al. 2019]	2019	EHR	not stated	no number in abstract
[Castro et al. 2020]	2020	Imaging	no dataset (review, theory, guidance or simulation)	-
[Kissas et al. 2020]	2020	Imaging	in-vivo 4D flow MRI measurements near the aorta/carotid bifurcation (a healthy human subject)	no number in abstract
[Lagergren et al. 2020]	2020	Tabular	wound healing assays with varying initial cell densities	no number in abstract
[Sahli Costabal et al. 2020]	2020	Signals	abstract not accessible	-
[Van Calster et al. 2020]	2020	Tabular	no dataset (review, theory, guidance or simulation)	-
[Köhler et al. 2021]	2021	Text	no dataset (review, theory, guidance or simulation)	-
[Lu et al. 2021a]	2021	Tabular	PK data from two treatment regimens of trastuzumab emtansine	no number in abstract
[Riley et al. 2021]	2021	Tabular	not stated	no number in abstract
[Zhang et al. 2021]	2021	Multimodal	eICU collaborative research database V2.0 (30,680 patients (study cohort)), over 200 sites; MIMIC-CXR; CheXpert; Chest-Xray8; PadChest	no number in abstract

P4 Unlabeled (18 works)

Work	Year	Modality	Data (size)	Reported result
[Agarwal et al. 2016]	2016	EHR	not stated	Precision/accuracy 0.90/0.89 (T2DM) and 0.86/0.89 (MI) vs rule-based 0.96/0.92 and 0.84/0.87
[Halpern et al. 2016]	2016	EHR	retrospective EMR data on emergency department patients	Triage-time AUCs: infection 0.89, cancer 0.88, septic shock 0.93, pneumonia 0.90; disposition-time AUCs up to 0.97 (septic shock, pneumonia)
[Bai et al. 2017]	2017	Imaging	abstract not accessible	-

Work	Year	Modality	Data (size)	Reported result
[Wang et al. 2017]	2017	Imaging	ChestX-ray8 (108,948 frontal-view X-ray images of 32,717 unique patients)	no number in abstract
[Ilse et al. 2018]	2018	Imaging	MNIST-bags (from MNIST); Breast cancer histopathology dataset (Gelasca et al., 2008) (58 H&E images); Colon cancer histopathology dataset (Sirinukunwattana et al., 2016) (100 H&E images)	no number in abstract
[Campanella et al. 2019]	2019	Imaging	whole slide image dataset (44,732 whole slide images from 15,187 patients)	AUC above 0.98 for prostate cancer, basal cell carcinoma and breast cancer metastases
[Cheplygina et al. 2019]	2019	Imaging	no dataset (review, theory, guidance or simulation)	-
[Fries et al. 2019]	2019	Imaging	UK Biobank (up to 4,000 unlabeled cardiac MRI sequences)	Model identifies individuals with a 1.8-fold increase in risk of a major adverse cardiac event; outperforms supervised hand-labeled model
[Irvin et al. 2019]	2019	Imaging	CheXpert (224,316 chest radiographs of 65,240 patients)	no number in abstract
[Wang et al. 2019]	2019	Text	Mayo Clinic smoking status classification (institutional), 1 site; Mayo Clinic proximal femur (hip) fracture classification (institutional), 1 site; i2b2 2006 smoking status classification shared task	CNN best: F1 0.92 (Mayo smoking status) and 0.97 (fracture classification).
[Yu et al. 2019]	2019	Imaging	MICCAI 2018 Atrial Segmentation Challenge dataset (100 3D gadolinium-enhanced MR imaging scans)	With 16 labeled scans, unlabeled data improves Dice by 2.85% and Jaccard 4.15% over labeled-only Bayesian V-Net
[Chaitanya et al. 2020]	2020	Imaging	ACDC (100 3D short-axis cardiac cine-MRIs); Prostate (MICCAI 2018 medical segmentation decathlon) (48 3D T2-weighted MRIs); MMWHS (20 3D cardiac MRIs)	Within 8% of benchmark using only two labeled MRI volumes (4% of ACDC training data)
[Ouyang et al. 2020]	2020	Imaging	not stated	no number in abstract
[Tajbakhsh et al. 2020]	2020	Imaging	no dataset (review, theory, guidance or simulation)	-
[Taleb et al. 2020]	2020	Imaging	3D MRI (Brain Tumor Segmentation); 3D CT (Pancreas Tumor Segmentation); 2D Fundus images (Diabetic Retinopathy Detection)	no number in abstract
[Li et al. 2021c]	2021	Imaging	International Skin Imaging Collaboration (ISIC) 2017; Retinal Fundus Glaucoma Challenge (REFUGE); Liver Tumor Segmentation Challenge (LiTS)	no number in abstract
[Lu et al. 2021b]	2021	Imaging	not stated	no number in abstract
[Krishnan et al. 2022]	2022	Multimodal	no dataset (review, theory, guidance or simulation)	-

P5 Procurement (12 works)

Work	Year	Modality	Data (size)	Reported result
[Peduzzi et al. 1996]	1996	Tabular	cardiac trial data (673 patients)	EPV of 10 or greater: no major problems; EPV below 10: biased coefficients and poor variance/coverage
[Figueroa et al. 2012]	2012	Text	learning curves from clinical text and waveform classification tasks	80 to 560 annotated samples needed for mean average and RMSE below 0.01; weighted fitting outperformed un-weighted baseline
[Cho et al. 2015]	2015	Imaging	Massachusetts General Hospital (MGH) PACS CT images (6000 test CT images; training sizes 5-200), 1 site	no number in abstract
[Yang et al. 2017]	2017	Imaging	2015 MICCAI Gland Challenge dataset; lymph node ultrasound image segmentation dataset	State-of-the-art segmentation performance achieved using only 50% of training data.
[Smailagic et al. 2018]	2018	Imaging	Messidor (1200 eye fundus images); Breast Cancer Diagnosis Dataset (ICIAR 2018 Grand Challenge) (400 images); Skin Cancer dataset (900 images)	80% DR detection accuracy with 425 labeled images; 32% fewer labels than uncertainty sampling, 40% fewer than random
[Balki et al. 2019]	2019	Imaging	no dataset (review, theory, guidance or simulation)	22 of 167 articles included; only 4 discussed sample-size determination methodologies
[Riley et al. 2019a]	2019	Tabular	no dataset (review, theory, guidance or simulation)	-
[Riley et al. 2019b]	2019	Tabular	no dataset (review, theory, guidance or simulation)	-

Work	Year	Modality	Data (size)	Reported result
[van Smeden et al. 2019]	2019	Tabular	no dataset (review, theory, guidance or simulation)	-
[Riley et al. 2020]	2020	Tabular	no dataset (review, theory, guidance or simulation)	-
[Budd et al. 2021]	2021	Imaging	no dataset (review, theory, guidance or simulation)	-
[Nath et al. 2021]	2021	Imaging	MRI scans of the hippocampus; CT scans of pancreas and tumors	Full accuracy using only 22.69% and 48.85% of available data for each dataset

P6 Re-expression (26 works)

Work	Year	Modality	Data (size)	Reported result
[Chawla et al. 2002]	2002	Tabular	not stated	no number in abstract
[Choi et al. 2017b]	2017	EHR	not stated	no number in abstract
[Bowles et al. 2018]	2018	Imaging	not stated	Dice Similarity Coefficient improvements of 1 to 5 percentage points from adding GAN-derived synthetic training data.
[Fernández et al. 2018]	2018	Tabular	no dataset (review, theory, guidance or simulation)	-
[Frid-Adar et al. 2018]	2018	Imaging	CT images of liver lesions (limited dataset) (182 liver lesions)	Sensitivity/specificity 85.7%/92.4% with synthetic augmentation vs 78.6%/88.4% with classic augmentation
[Shin et al. 2018]	2018	Imaging	Alzheimer's Disease Neuroimaging Initiative (ADNI) (3,416 pairs of T1-weighted images); BRATS 2015 training data set (264 studies)	no number in abstract
[Walonoski et al. 2018]	2018	EHR	Synthea synthetic patient records (One million synthetic patient records)	no number in abstract
[Baowaly et al. 2019]	2019	EHR	MIMIC-III; National Health Insurance Research Database (NHIRD), Taiwan	no number in abstract
[Beaulieu-Jones et al. 2019]	2019	Tabular	SPRINT trial (Systolic Blood Pressure Trial)	no number in abstract
[Jordon et al. 2019]	2019	Tabular	abstract not accessible	-
[Xu et al. 2019]	2019	Tabular	adult (UCI); census (UCI); covertype (UCI); intrusion (UCI); news (UCI); credit (Kaggle); MNIST28 / MNIST12 (from MNIST); 7 simulated datasets	CTGAN surpasses Bayesian networks on at least 87.5% of benchmark datasets
[Yi et al. 2019]	2019	Imaging	no dataset (review, theory, guidance or simulation)	-
[Zhao et al. 2019]	2019	Imaging	not stated	no number in abstract
[Goncalves et al. 2020]	2020	EHR	Surveillance Epidemiology and End Results (SEER) cancer registry (over 360,000 individual cases)	no number in abstract
[Yoon et al. 2020]	2020	EHR	four independent datasets (4 datasets)	no number in abstract
[Zhang et al. 2020]	2020	EHR	billing codes from EMRs at Vanderbilt University Medical Center (over 1 million EMRs), 1 site	no number in abstract
[Chaitanya et al. 2021]	2021	Imaging	cardiac; prostate; pancreas	no number in abstract
[Chen et al. 2021a]	2021	Multimodal	abstract not accessible	-
[Chlap et al. 2021]	2021	Imaging	no dataset (review, theory, guidance or simulation)	-
[Stadler et al. 2022]	2022	Tabular	not stated	no number in abstract
[van den Goorbergh et al. 2022]	2022	Tabular	ovarian cancer diagnosis case study data	no number in abstract
[Borisov et al. 2023]	2023	Tabular	numerous real-world and synthetic data sets (not named)	no number in abstract
[Kazerouni et al. 2023]	2023	Imaging	no dataset (review, theory, guidance or simulation)	-
[Khader et al. 2023]	2023	Imaging	ADNI (brain MRI) (998 studies); LIDC (chest CT) (1010 studies); DUKE (breast MRI) (1844 studies); MRNet (knee MRI) (1250 studies)	Breast segmentation Dice 0.95 with synthetic data vs 0.91 without
[Kotelnikov et al. 2023]	2023	Tabular	not stated	no number in abstract
[Ktena et al. 2024]	2024	Imaging	histopathology images; chest X-ray images; dermatology images	no number in abstract

No remedy (documents a problem) (26 works)

Work	Year	Modality	Data (size)	Reported result
[El Emam et al. 2011]	2011	Tabular	no dataset (review, theory, guidance or simulation)	Pooled re-identified proportion 0.26 (95% CI 0.046-0.478) overall and 0.34 for health data, across 14 attacks.
[Varoquaux 2018]	2018	Imaging	not stated	Sample sizes of many neuroimaging studies lead to large cross-validation error bars, e.g. ±10% for 100 samples.
[Zech et al. 2018]	2018	Imaging	National Institutes of Health Clinical Center (NIH) (112,120 radiographs from 30,805 patients), 1 site; Mount Sinai Hospital (MSH) (42,396 radiographs from 12,904 patients), 1 site; Indiana University Network for Patient Care (IU) (3,807 radiographs from 3,683 patients), 1 site	Sorting by hospital system alone achieved AUC 0.861 on joint MSH-NIH data; better internal than external in 3/5 comparisons
[Brasil et al. 2019]	2019	Multimodal	no dataset (review, theory, guidance or simulation)	-
[Christodoulou et al. 2019]	2019	Tabular	no dataset (review, theory, guidance or simulation)	Low-risk-of-bias comparisons (145): logit(AUC) difference LR vs ML 0.00 (95% CI -0.18 to 0.18)
[Kim et al. 2019]	2019	Imaging	no dataset (review, theory, guidance or simulation)	Of 516 eligible studies, only 6% (31 studies) performed external validation
[Liu et al. 2019]	2019	Imaging	no dataset (review, theory, guidance or simulation)	Pooled sensitivity 87.0% (DL) vs 86.4% (clinicians); pooled specificity 92.5% vs 90.5%, in 14 studies.
[Obermeyer et al. 2019]	2019	EHR	not stated	Remedying the bias would increase Black patients receiving additional help from 17.7% to 46.5%.
[Rocher et al. 2019]	2019	Tabular	210 populations (unnamed) (210 populations)	AUC 0.84-0.97 for predicting individual uniqueness; 99.98% of Americans re-identifiable with 15 attributes
[Schwarz et al. 2019]	2019	Imaging	deidentified cranial MRI scans	no number in abstract
[Nagendran et al. 2020]	2020	Imaging	no dataset (review, theory, guidance or simulation)	High overall risk of bias in 58 of 81 non-randomised studies; data and code unavailable in 95% and 93%
[Nguengang Wakap et al. 2020]	2020	Tabular	Orphanet database (6172 unique rare diseases)	Population prevalence of rare diseases estimated at 3.5-5.9%, i.e. 263-446 million persons globally
[Oakden-Rayner et al. 2020]	2020	Imaging	CIFAR-100; multiple real-world medical imaging datasets	Hidden stratification can cause relative performance differences of over 20% on clinically important subsets.
[Schaefer et al. 2020]	2020	Multimodal	no dataset (review, theory, guidance or simulation)	211 studies from 32 countries on 74 rare diseases; only 11.8% evaluated on an external data set.
[Willemink et al. 2020]	2020	Imaging	no dataset (review, theory, guidance or simulation)	-
[Wynants et al. 2020]	2020	Multimodal	no dataset (review, theory, guidance or simulation)	169 studies, 232 models; all at high or unclear risk of bias; C index 0.54 to >0.99
[Chen et al. 2021b]	2021	Multimodal	no dataset (review, theory, guidance or simulation)	-
[Finlayson et al. 2021]	2021	Multimodal	no dataset (review, theory, guidance or simulation)	-
[Koh et al. 2021]	2021	Multimodal	WILDS benchmark (10 datasets)	no number in abstract
[Roberts et al. 2021]	2021	Imaging	no dataset (review, theory, guidance or simulation)	61 studies included (of 2,212 identified); none of the models identified are of potential clinical use.
[Wong et al. 2021]	2021	EHR	Michigan Medicine (University of Michigan) hospitalized patients (27,697 patients with 38,455 hospitalizations), 1 site	Epic Sepsis Model hospitalization-level AUROC 0.63 (95% CI 0.62-0.64)
[Gichoya et al. 2022]	2022	Imaging	Emory CXR; Emory Chest CT; Emory Cervical Spine; Emory Mammogram; MIMIC-CXR; CheXpert; National Lung Cancer Screening Trial; RSNA Pulmonary Embolism CT; Digital Hand Atlas	Race prediction AUC 0·91-0·99 (x-ray), 0·87-0·96 (CT chest), 0·81 (mammography) under external validation
[Packhäuser et al. 2022]	2022	Imaging	ChestX-ray14 (112,120 frontal-view chest X-ray images from 30,805 unique patients); CheXpert; COVID-19 Image Data Collection	Same-patient verification AUC 0.9940, accuracy 95.55% on ChestX-ray14; external datasets AUC up to 0.9870
[Varoquaux and Cheplygina 2022]	2022	Imaging	no dataset (review, theory, guidance or simulation)	-
[Yu et al. 2022]	2022	Imaging	no dataset (review, theory, guidance or simulation)	81% (70 of 86) algorithms showed decreased external vs internal performance; 24% a substantial decrease (>=0.10).

Work	Year	Modality	Data (size)	Reported result
[Hatamizadeh et al. 2023]	2023	Imaging	not stated	no number in abstract

S4 DOES THE SMALLNESS INDEX PREDICT THE GAIN FROM BORROWING?

The test of Section 2.5 of the paper followed a plan written and committed before any correlation was computed (`simulations/dev/SIGMA_PLAN.md`; code `simulations/sigma_validation.py`; output `supplement/results/sigma_validation.json`). Hospitals: the five held-out TCGA networks (GHB as released), the twelve TCGA design networks (the same configuration), and Arizona Medicare and UCI Heart Disease, rerun with the folds of the seven-network study, local estimation and GHB only, and per-hospital losses kept. The six small TCGA networks, whose hospitals largely coincide with six design networks, form a secondary set (Spearman correlation of σ with the gain -0.10 at $\epsilon = 1$ and $+0.08$ at $\epsilon = 10$, of n $+0.01$ and -0.18). Gain: one minus the ratio of summed held-out log-losses of GHB and local estimation. $n^\star$: the largest of the three criteria of Riley et al. for a binary outcome (shrinkage 0.9, optimism in Nagelkerke R^2 at most 0.05, intercept within 0.05), with p the number of predictor parameters, prevalence clipped to $[0.5/n, 1 - 0.5/n]$ and anticipated R^2_{CS} equal to 15% of its maximum; our implementation reproduces the $n^\star = 2{,}707$ of Example 2.4. The claim rule was that σ is a better predictor only if the lower end of the 95% cluster-bootstrap interval of D is above zero at both budgets. It is below zero at both, and σ is reported in the paper as a diagnostic, not as a better predictor. Table S11 lists every hospital.

Table S11. Every hospital of the test: size, events, events per variable (EPV), Riley minimum sample size $n^\star$, smallness index σ and gain from GHB over local estimation (one minus the ratio of held-out log-losses). Sets: held out and design TCGA networks, and the networks of the seven-network study (small; Arizona Medicare and UCI Heart Disease are in the primary set, the six TCGA networks in the secondary set).

Network	Set	n	Events	EPV	$n^\star$	σ	Gain, $\epsilon = 1$	Gain, $\epsilon = 10$
TCGA-BLCA	small	44	21	5.2	383	8.71	+0.001	+0.026
TCGA-BLCA	small	32	10	2.5	330	10.32	+0.086	+0.123
TCGA-BLCA	small	29	13	3.2	380	13.11	+0.090	+0.114
TCGA-BLCA	small	27	10	2.5	358	13.27	+0.032	+0.052
TCGA-BLCA	small	21	4	1.0	365	17.39	−0.041	−0.015
TCGA-BLCA	small	12	4	1.0	341	28.46	+0.001	+0.117
TCGA-BLCA	small	11	3	0.8	327	29.76	+0.048	+0.126
TCGA-BLCA	small	10	4	1.0	369	36.88	+0.085	+0.142
TCGA-BLCA	small	10	4	1.0	369	36.88	+0.090	+0.126
TCGA-BLCA	small	7	0	0.1	576	82.34	−0.762	−1.013
TCGA-BLCA	small	6	2	0.5	341	56.91	+0.264	+0.364
TCGA-BLCA	small	6	1	0.2	384	63.96	+0.129	+0.214
TCGA-BLCA	small	5	1	0.2	359	71.83	−0.070	+0.060
TCGA-HNSC	small	116	57	14.3	384	3.31	+0.023	+0.025
TCGA-HNSC	small	52	24	6.0	382	7.34	+0.054	+0.051
TCGA-HNSC	small	33	12	3.0	356	10.78	+0.020	+0.040
TCGA-HNSC	small	24	6	1.5	335	13.96	+0.178	+0.203
TCGA-HNSC	small	17	6	1.5	351	20.64	+0.103	+0.124
TCGA-HNSC	small	13	6	1.5	382	29.38	+0.059	+0.083
TCGA-HNSC	small	9	4	1.0	379	42.16	+0.259	+0.338
TCGA-HNSC	small	7	3	0.8	376	53.76	+0.005	+0.117
TCGA-HNSC	small	6	2	0.5	341	56.91	+0.184	+0.282
TCGA-HNSC	small	6	1	0.2	384	63.96	+0.043	+0.119
TCGA-HNSC	small	6	3	0.8	384	64.03	−0.076	−0.065
TCGA-HNSC	small	5	2	0.5	369	73.76	−0.336	−0.477
TCGA-KIRC	small	94	13	3.2	414	4.41	−0.025	−0.032
TCGA-KIRC	small	70	32	8.0	381	5.45	−0.009	+0.009
TCGA-KIRC	small	54	11	2.8	357	6.61	+0.044	+0.076
TCGA-KIRC	small	29	10	2.5	347	11.97	−0.044	+0.030
TCGA-KIRC	small	23	2	0.5	518	22.51	+0.053	+0.117
TCGA-KIRC	small	10	3	0.8	323	32.27	−0.017	−0.027
TCGA-KIRC	small	9	2	0.5	347	38.55	−0.101	−0.023
TCGA-LIHC	small	101	30	7.5	321	3.18	+0.034	+0.034

Table S11. Every hospital of the test: size, events, events per variable (EPV), Riley minimum sample size $n^\star$, smallness index σ and gain from GHB over local estimation (one minus the ratio of held-out log-losses). Sets: held out and design TCGA networks, and the networks of the seven-network study (small; Arizona Medicare and UCI Heart Disease are in the primary set, the six TCGA networks in the secondary set).

Network	Set	n	Events	EPV	$n^\star$	σ	Gain, $\epsilon = 1$	Gain, $\epsilon = 10$
TCGA-LIHC	small	22	0	0.1	1211	55.02	−0.440	−0.602
TCGA-LIHC	small	13	2	0.5	396	30.47	+0.142	+0.181
TCGA-LIHC	small	12	6	1.5	384	32.01	+0.247	+0.268
TCGA-LIHC	small	7	3	0.8	376	53.76	+0.140	+0.147
TCGA-LIHC	small	7	2	0.5	324	46.22	+0.158	+0.183
TCGA-LIHC	small	5	1	0.2	359	71.83	−0.190	−0.159
TCGA-LUSC	small	43	12	3.0	325	7.57	+0.073	+0.076
TCGA-LUSC	small	36	13	3.2	355	9.85	−0.015	−0.034
TCGA-LUSC	small	35	17	4.2	384	10.97	−0.041	−0.033
TCGA-LUSC	small	22	6	1.5	327	14.88	+0.082	+0.086
TCGA-LUSC	small	21	6	1.5	324	15.41	−0.096	−0.140
TCGA-LUSC	small	20	6	1.5	323	16.13	+0.107	+0.128
TCGA-LUSC	small	18	7	1.8	365	20.29	+0.107	+0.109
TCGA-LUSC	small	18	3	0.8	384	21.32	+0.119	+0.159
TCGA-LUSC	small	17	4	1.0	341	20.06	+0.022	+0.045
TCGA-LUSC	small	17	8	2.0	383	22.52	+0.056	+0.081
TCGA-LUSC	small	15	5	1.3	341	22.77	+0.218	+0.270
TCGA-LUSC	small	14	2	0.5	408	29.18	−0.299	−0.487
TCGA-LUSC	small	14	5	1.2	353	25.20	+0.091	+0.124
TCGA-LUSC	small	12	5	1.2	373	31.12	+0.081	+0.091
TCGA-LUSC	small	12	4	1.0	341	28.46	−0.029	−0.042
TCGA-LUSC	small	8	4	1.0	384	48.02	+0.072	+0.104
TCGA-LUSC	small	6	2	0.5	341	56.91	−0.058	−0.191
TCGA-STAD	small	81	37	9.2	381	4.71	+0.015	+0.027
TCGA-STAD	small	63	28	7.0	379	6.02	−0.019	+0.007
TCGA-STAD	small	23	9	2.2	366	15.91	+0.067	+0.099
TCGA-STAD	small	14	3	0.8	351	25.07	+0.200	+0.279
TCGA-STAD	small	14	6	1.5	376	26.88	+0.175	+0.243
TCGA-STAD	small	12	3	0.8	335	27.92	+0.141	+0.153
TCGA-STAD	small	10	1	0.1	712	71.22	−0.350	−0.467
TCGA-STAD	small	7	2	0.5	324	46.22	−0.057	−0.005
TCGA-STAD	small	5	1	0.2	359	71.83	−0.042	+0.108
heart	small	303	139	34.8	382	1.26	−0.003	−0.002
heart	small	294	106	26.5	354	1.21	+0.004	+0.009
heart	small	123	8	2.0	608	4.94	+0.035	+0.044
heart	small	200	51	12.8	333	1.67	+0.007	+0.013
medpar	small	92	38	9.5	373	4.05	+0.021	+0.030
medpar	small	74	23	5.8	329	4.45	+0.058	+0.069
medpar	small	71	32	8.0	380	5.36	+0.020	+0.022
medpar	small	66	18	4.5	327	4.96	+0.043	+0.064
medpar	small	64	14	3.5	349	5.45	+0.062	+0.063
medpar	small	61	19	4.8	330	5.40	+0.052	+0.066
medpar	small	61	19	4.8	330	5.40	+0.038	+0.052
medpar	small	58	16	4.0	326	5.63	+0.015	+0.018
medpar	small	55	22	5.5	369	6.71	+0.053	+0.077
medpar	small	51	10	2.5	362	7.09	+0.028	+0.024
medpar	small	50	18	4.5	354	7.08	+0.017	+0.019
medpar	small	44	17	4.2	364	8.28	+0.088	+0.105
medpar	small	43	17	4.2	367	8.54	+0.081	+0.098
medpar	small	41	15	3.8	357	8.70	+0.019	+0.034
medpar	small	39	12	3.0	327	8.39	+0.076	+0.101
medpar	small	38	14	3.5	358	9.41	+0.053	+0.081
medpar	small	38	18	4.5	383	10.08	−0.046	−0.061
medpar	small	36	12	3.0	341	9.49	+0.022	+0.034
medpar	small	33	10	2.5	325	9.83	+0.053	+0.067
medpar	small	31	9	2.2	322	10.40	+0.042	+0.050
medpar	small	31	11	2.8	352	11.35	+0.047	+0.065
medpar	small	29	13	3.2	380	13.11	+0.008	+0.006
medpar	small	29	13	3.2	380	13.11	+0.035	+0.026
medpar	small	28	11	2.8	367	13.09	−0.010	−0.019
medpar	small	27	11	2.8	371	13.74	+0.086	+0.132
medpar	small	26	9	2.2	348	13.38	+0.031	+0.023
medpar	small	25	5	1.2	359	14.37	+0.119	+0.113

Table S11. Every hospital of the test: size, events, events per variable (EPV), Riley minimum sample size $n^\star$, smallness index σ and gain from GHB over local estimation (one minus the ratio of held-out log-losses). Sets: held out and design TCGA networks, and the networks of the seven-network study (small; Arizona Medicare and UCI Heart Disease are in the primary set, the six TCGA networks in the secondary set).

Network	Set	n	Events	EPV	$n^\star$	σ	Gain, $\epsilon = 1$	Gain, $\epsilon = 10$
medpar	small	21	9	2.2	376	17.92	+0.108	+0.117
medpar	small	21	7	1.8	341	16.26	+0.094	+0.119
medpar	small	20	3	0.8	400	20.01	+0.170	+0.186
medpar	small	19	8	2.0	375	19.71	+0.072	+0.073
medpar	small	18	7	1.8	365	20.29	+0.007	−0.003
medpar	small	17	5	1.2	321	18.90	−0.020	−0.045
medpar	small	16	4	1.0	335	20.94	−0.001	−0.011
medpar	small	15	1	0.2	599	39.96	+0.191	+0.226
medpar	small	15	3	0.8	359	23.94	+0.237	+0.281
medpar	small	13	2	0.5	396	30.47	−0.222	−0.263
medpar	small	12	3	0.8	335	27.92	+0.010	+0.044
medpar	small	11	4	1.0	356	32.33	−0.004	+0.002
medpar	small	10	5	1.2	384	38.42	−0.151	−0.162
medpar	small	6	2	0.5	341	56.91	+0.009	+0.008
medpar	small	6	3	0.8	384	64.03	+0.074	+0.086
medpar	small	5	1	0.2	359	71.83	+0.201	+0.227
TCGA-BRCA	held out	47	9	1.3	638	13.57	−0.017	−0.017
TCGA-BRCA	held out	29	6	0.9	621	21.43	+0.187	+0.208
TCGA-BRCA	held out	15	4	0.6	576	38.41	+0.257	+0.287
TCGA-BRCA	held out	10	4	0.6	532	53.20	+0.097	+0.161
TCGA-BRCA	held out	6	2	0.3	548	91.26	−0.207	−0.194
TCGA-BRCA	held out	6	2	0.3	548	91.26	+0.305	+0.327
TCGA-BRCA	held out	6	3	0.4	524	87.37	−0.063	−0.123
TCGA-BRCA	held out	6	1	0.1	672	111.93	+0.158	+0.215
TCGA-MESO	held out	14	5	0.7	541	38.63	−0.062	−0.052
TCGA-MESO	held out	12	5	0.7	530	44.13	+0.190	+0.208
TCGA-MESO	held out	9	1	0.1	801	88.97	+0.081	+0.045
TCGA-MESO	held out	8	3	0.4	537	67.09	+0.020	+0.027
TCGA-MESO	held out	7	2	0.3	566	80.89	−0.032	−0.101
TCGA-MESO	held out	7	3	0.4	528	75.45	−0.137	−0.069
TCGA-MESO	held out	5	2	0.3	532	106.40	+0.148	+0.181
TCGA-MESO	held out	5	2	0.3	532	106.40	+0.300	+0.340
TCGA-MESO	held out	5	1	0.1	628	125.69	+0.198	+0.222
TCGA-OV	held out	109	17	2.4	689	6.32	−0.010	+0.004
TCGA-OV	held out	73	22	3.1	559	7.66	−0.017	−0.017
TCGA-OV	held out	48	13	1.9	574	11.95	+0.042	+0.046
TCGA-OV	held out	45	18	2.6	532	11.82	+0.009	+0.025
TCGA-OV	held out	39	7	1.0	653	16.75	+0.043	+0.066
TCGA-OV	held out	35	6	0.9	664	18.98	−0.026	+0.006
TCGA-OV	held out	27	4	0.6	704	26.08	−0.193	−0.310
TCGA-OV	held out	23	5	0.7	611	26.58	−0.029	−0.029
TCGA-OV	held out	16	5	0.7	555	34.67	+0.093	+0.109
TCGA-OV	held out	15	2	0.3	736	49.10	+0.154	+0.192
TCGA-OV	held out	12	5	0.7	530	44.13	+0.090	+0.116
TCGA-OV	held out	9	4	0.6	527	58.51	+0.055	+0.065
TCGA-OV	held out	6	1	0.1	672	111.93	+0.134	+0.128
TCGA-OV	held out	6	2	0.3	548	91.26	+0.222	+0.326
TCGA-OV	held out	5	2	0.3	532	106.40	+0.047	+0.047
TCGA-UCEC	held out	51	2	0.3	1456	28.54	+0.140	+0.177
TCGA-UCEC	held out	51	5	0.7	852	16.70	−0.084	−0.108
TCGA-UCEC	held out	42	5	0.7	775	18.45	+0.124	+0.158
TCGA-UCEC	held out	37	1	0.1	1874	50.64	+0.095	+0.116
TCGA-UCEC	held out	34	8	1.1	597	17.55	−0.023	+0.003
TCGA-UCEC	held out	25	1	0.1	1437	57.47	+0.076	+0.067
TCGA-UCEC	held out	21	3	0.4	715	34.04	+0.071	+0.104
TCGA-UCEC	held out	20	7	1.0	543	27.14	−0.004	+0.016
TCGA-UCEC	held out	19	1	0.1	1208	63.55	−0.016	−0.004
TCGA-UCEC	held out	15	2	0.3	736	49.10	−0.083	−0.065
TCGA-UCEC	held out	14	7	1.0	524	37.44	+0.181	+0.257
TCGA-UCEC	held out	8	2	0.3	586	73.29	−0.078	−0.046
TCGA-UCEC	held out	8	0	0.1	1089	136.15	−0.196	−0.135
TCGA-UCEC	held out	5	1	0.1	628	125.69	−0.104	−0.137
TCGA-UCEC	held out	5	0	0.1	843	168.62	−0.123	−0.096

Table S11. Every hospital of the test: size, events, events per variable (EPV), Riley minimum sample size $n^\star$, smallness index σ and gain from GHB over local estimation (one minus the ratio of held-out log-losses). Sets: held out and design TCGA networks, and the networks of the seven-network study (small; Arizona Medicare and UCI Heart Disease are in the primary set, the six TCGA networks in the secondary set).

Network	Set	n	Events	EPV	$n^\star$	σ	Gain, $\epsilon = 1$	Gain, $\epsilon = 10$
TCGA-UCS	held out	14	6	0.9	528	37.72	−0.005	−0.054
TCGA-UCS	held out	7	3	0.4	528	75.45	+0.050	+0.109
TCGA-UCS	held out	7	3	0.4	528	75.45	+0.272	+0.326
TCGA-UCS	held out	6	2	0.3	548	91.26	+0.017	−0.048
TCGA-UCS	held out	5	2	0.3	532	106.40	+0.161	+0.231
TCGA-BLCA	design	44	21	3.0	525	11.92	+0.055	+0.064
TCGA-BLCA	design	32	10	1.4	555	17.34	+0.164	+0.206
TCGA-BLCA	design	29	13	1.9	526	18.15	+0.189	+0.208
TCGA-BLCA	design	27	10	1.4	538	19.91	+0.148	+0.159
TCGA-BLCA	design	21	4	0.6	639	30.44	−0.015	−0.020
TCGA-BLCA	design	12	4	0.6	548	45.63	+0.045	+0.108
TCGA-BLCA	design	11	3	0.4	573	52.07	+0.136	+0.199
TCGA-BLCA	design	10	4	0.6	532	53.20	+0.195	+0.224
TCGA-BLCA	design	10	4	0.6	532	53.20	+0.193	+0.165
TCGA-BLCA	design	7	0	0.1	1009	144.09	−1.252	−1.462
TCGA-BLCA	design	6	2	0.3	548	91.26	+0.340	+0.415
TCGA-BLCA	design	6	1	0.1	672	111.93	+0.220	+0.213
TCGA-BLCA	design	5	1	0.1	628	125.69	+0.059	+0.088
TCGA-COAD	design	101	27	3.9	576	5.70	+0.006	+0.007
TCGA-COAD	design	40	4	0.6	843	21.08	+0.158	+0.153
TCGA-COAD	design	24	6	0.9	586	24.43	+0.087	+0.120
TCGA-COAD	design	23	3	0.4	744	32.33	+0.031	+0.061
TCGA-COAD	design	20	10	1.4	524	26.21	+0.031	+0.092
TCGA-COAD	design	14	4	0.6	566	40.44	+0.086	+0.154
TCGA-COAD	design	9	1	0.1	801	88.97	+0.211	+0.371
TCGA-COAD	design	7	1	0.1	715	102.12	−0.032	−0.155
TCGA-COAD	design	5	2	0.3	532	106.40	+0.303	+0.343
TCGA-COAD	design	5	1	0.1	628	125.69	+0.252	+0.276
TCGA-COAD	design	5	1	0.1	628	125.69	+0.309	+0.398
TCGA-ESCA	design	27	12	1.7	527	19.50	+0.114	+0.152
TCGA-ESCA	design	14	6	0.9	528	37.72	+0.145	+0.161
TCGA-ESCA	design	12	4	0.6	548	45.63	+0.071	+0.123
TCGA-ESCA	design	11	1	0.1	885	80.46	+0.279	+0.322
TCGA-ESCA	design	10	2	0.3	628	62.85	+0.047	+0.116
TCGA-ESCA	design	6	1	0.1	672	111.93	+0.076	+0.153
TCGA-HNSC	design	116	57	8.1	524	4.52	+0.028	+0.026
TCGA-HNSC	design	52	24	3.4	525	10.10	+0.121	+0.113
TCGA-HNSC	design	33	12	1.7	539	16.34	+0.103	+0.118
TCGA-HNSC	design	24	6	0.9	586	24.43	+0.328	+0.335
TCGA-HNSC	design	17	6	0.9	542	31.88	+0.123	+0.125
TCGA-HNSC	design	13	6	0.9	525	40.41	+0.197	+0.223
TCGA-HNSC	design	9	4	0.6	527	58.51	+0.397	+0.430
TCGA-HNSC	design	7	3	0.4	528	75.45	+0.150	+0.238
TCGA-HNSC	design	6	2	0.3	548	91.26	+0.185	+0.256
TCGA-HNSC	design	6	1	0.1	672	111.93	−0.027	+0.006
TCGA-HNSC	design	6	3	0.4	524	87.37	−0.256	−0.302
TCGA-HNSC	design	5	2	0.3	532	106.40	−0.332	−0.210
TCGA-KIRC	design	94	13	1.9	725	7.71	−0.007	+0.005
TCGA-KIRC	design	70	32	4.6	526	7.51	−0.011	−0.002
TCGA-KIRC	design	54	11	1.6	625	11.57	+0.139	+0.187
TCGA-KIRC	design	29	10	1.4	544	18.76	−0.006	+0.011
TCGA-KIRC	design	23	2	0.3	906	39.39	+0.140	+0.214
TCGA-KIRC	design	10	3	0.4	560	55.98	−0.164	−0.259
TCGA-KIRC	design	9	2	0.3	607	67.46	−0.062	+0.055
TCGA-KIRP	design	24	6	0.9	586	24.43	−0.010	+0.026
TCGA-KIRP	design	21	2	0.3	864	41.15	+0.112	+0.169
TCGA-KIRP	design	14	0	0.1	1548	110.60	−0.418	−0.416
TCGA-KIRP	design	13	2	0.3	693	53.33	+0.006	+0.165
TCGA-KIRP	design	12	4	0.6	548	45.63	−0.564	−0.614
TCGA-KIRP	design	10	2	0.3	628	62.85	+0.025	+0.127
TCGA-KIRP	design	7	3	0.4	528	75.45	+0.138	+0.164
TCGA-KIRP	design	6	0	0.1	927	154.44	−0.547	−0.290
TCGA-KIRP	design	6	0	0.1	927	154.44	−0.757	−0.604

Table S11. Every hospital of the test: size, events, events per variable (EPV), Riley minimum sample size $n^\star$, smallness index σ and gain from GHB over local estimation (one minus the ratio of held-out log-losses). Sets: held out and design TCGA networks, and the networks of the seven-network study (small; Arizona Medicare and UCI Heart Disease are in the primary set, the six TCGA networks in the secondary set).

Network	Set	n	Events	EPV	$n^\star$	σ	Gain, $\epsilon = 1$	Gain, $\epsilon = 10$
TCGA-KIRP	design	6	1	0.1	672	111.93	+0.277	+0.318
TCGA-KIRP	design	5	0	0.1	843	168.62	−0.168	−0.215
TCGA-LIHC	design	101	30	4.3	561	5.55	+0.046	+0.054
TCGA-LIHC	design	22	0	0.1	2118	96.29	−0.319	−0.407
TCGA-LIHC	design	13	2	0.3	693	53.33	+0.186	+0.206
TCGA-LIHC	design	12	6	0.9	524	43.69	+0.346	+0.380
TCGA-LIHC	design	7	3	0.4	528	75.45	+0.233	+0.208
TCGA-LIHC	design	7	2	0.3	566	80.89	+0.210	+0.217
TCGA-LIHC	design	5	1	0.1	628	125.69	−0.196	−0.226
TCGA-LUAD	design	29	9	1.3	556	19.16	+0.186	+0.180
TCGA-LUAD	design	26	12	1.7	525	20.21	−0.037	−0.014
TCGA-LUAD	design	21	7	1.0	548	26.07	+0.029	−0.007
TCGA-LUAD	design	17	8	1.1	525	30.87	+0.261	+0.286
TCGA-LUAD	design	17	6	0.9	542	31.88	+0.087	+0.110
TCGA-LUAD	design	9	4	0.6	527	58.51	+0.051	+0.048
TCGA-LUAD	design	9	1	0.1	801	88.97	−0.096	−0.036
TCGA-LUAD	design	9	4	0.6	527	58.51	+0.141	+0.120
TCGA-LUAD	design	9	2	0.3	607	67.46	+0.356	+0.333
TCGA-LUAD	design	7	1	0.1	715	102.12	+0.073	−0.010
TCGA-LUAD	design	6	3	0.4	524	87.37	+0.165	+0.139
TCGA-LUSC	design	43	12	1.7	569	13.24	+0.113	+0.114
TCGA-LUSC	design	36	13	1.9	540	15.00	−0.007	−0.027
TCGA-LUSC	design	35	17	2.4	524	14.98	−0.031	−0.041
TCGA-LUSC	design	22	6	0.9	573	26.04	+0.192	+0.201
TCGA-LUSC	design	21	6	0.9	566	26.96	−0.036	−0.045
TCGA-LUSC	design	20	6	0.9	560	27.99	+0.077	+0.073
TCGA-LUSC	design	18	7	1.0	534	29.66	+0.317	+0.344
TCGA-LUSC	design	18	3	0.4	672	37.31	+0.162	+0.202
TCGA-LUSC	design	17	4	0.6	597	35.10	+0.120	+0.144
TCGA-LUSC	design	17	8	1.1	525	30.87	+0.180	+0.202
TCGA-LUSC	design	15	5	0.7	548	36.50	+0.299	+0.314
TCGA-LUSC	design	14	2	0.3	715	51.06	−0.205	−0.197
TCGA-LUSC	design	14	5	0.7	541	38.63	−0.126	−0.109
TCGA-LUSC	design	12	5	0.7	530	44.13	+0.259	+0.249
TCGA-LUSC	design	12	4	0.6	548	45.63	−0.083	−0.095
TCGA-LUSC	design	8	4	0.6	524	65.53	+0.020	+0.007
TCGA-LUSC	design	6	2	0.3	548	91.26	−0.026	−0.143
TCGA-PAAD	design	26	5	0.7	637	24.50	−0.025	+0.019
TCGA-PAAD	design	17	2	0.3	779	45.85	+0.207	+0.205
TCGA-PAAD	design	15	4	0.6	576	38.41	+0.079	+0.100
TCGA-PAAD	design	15	4	0.6	576	38.41	+0.007	+0.008
TCGA-PAAD	design	9	2	0.3	607	67.46	−0.227	−0.154
TCGA-PAAD	design	7	2	0.3	566	80.89	+0.315	+0.353
TCGA-PAAD	design	6	1	0.1	672	111.93	+0.173	+0.215
TCGA-PAAD	design	6	1	0.1	672	111.93	−0.044	−0.007
TCGA-PAAD	design	5	0	0.1	843	168.62	−1.187	−1.129
TCGA-PAAD	design	5	2	0.3	532	106.40	+0.355	+0.278
TCGA-READ	design	35	6	0.9	664	18.98	−0.001	+0.041
TCGA-READ	design	15	3	0.4	628	41.90	+0.106	+0.112
TCGA-READ	design	10	3	0.4	560	55.98	−0.711	−0.894
TCGA-READ	design	10	1	0.1	843	84.31	+0.061	+0.043
TCGA-READ	design	7	0	0.1	1009	144.09	−0.466	−0.532
TCGA-STAD	design	81	37	5.3	526	6.49	+0.010	+0.013
TCGA-STAD	design	63	28	4.0	527	8.36	−0.009	+0.002
TCGA-STAD	design	23	9	1.3	533	23.20	+0.213	+0.270
TCGA-STAD	design	14	3	0.4	614	43.87	+0.309	+0.357
TCGA-STAD	design	14	6	0.9	528	37.72	+0.288	+0.320
TCGA-STAD	design	12	3	0.4	586	48.86	+0.190	+0.206
TCGA-STAD	design	10	1	0.1	1246	124.64	−0.552	−0.569
TCGA-STAD	design	7	2	0.3	566	80.89	+0.031	+0.008
TCGA-STAD	design	5	1	0.1	628	125.69	+0.058	+0.179

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